

Fast prototyping of automated robotic platforms for critical materials separation

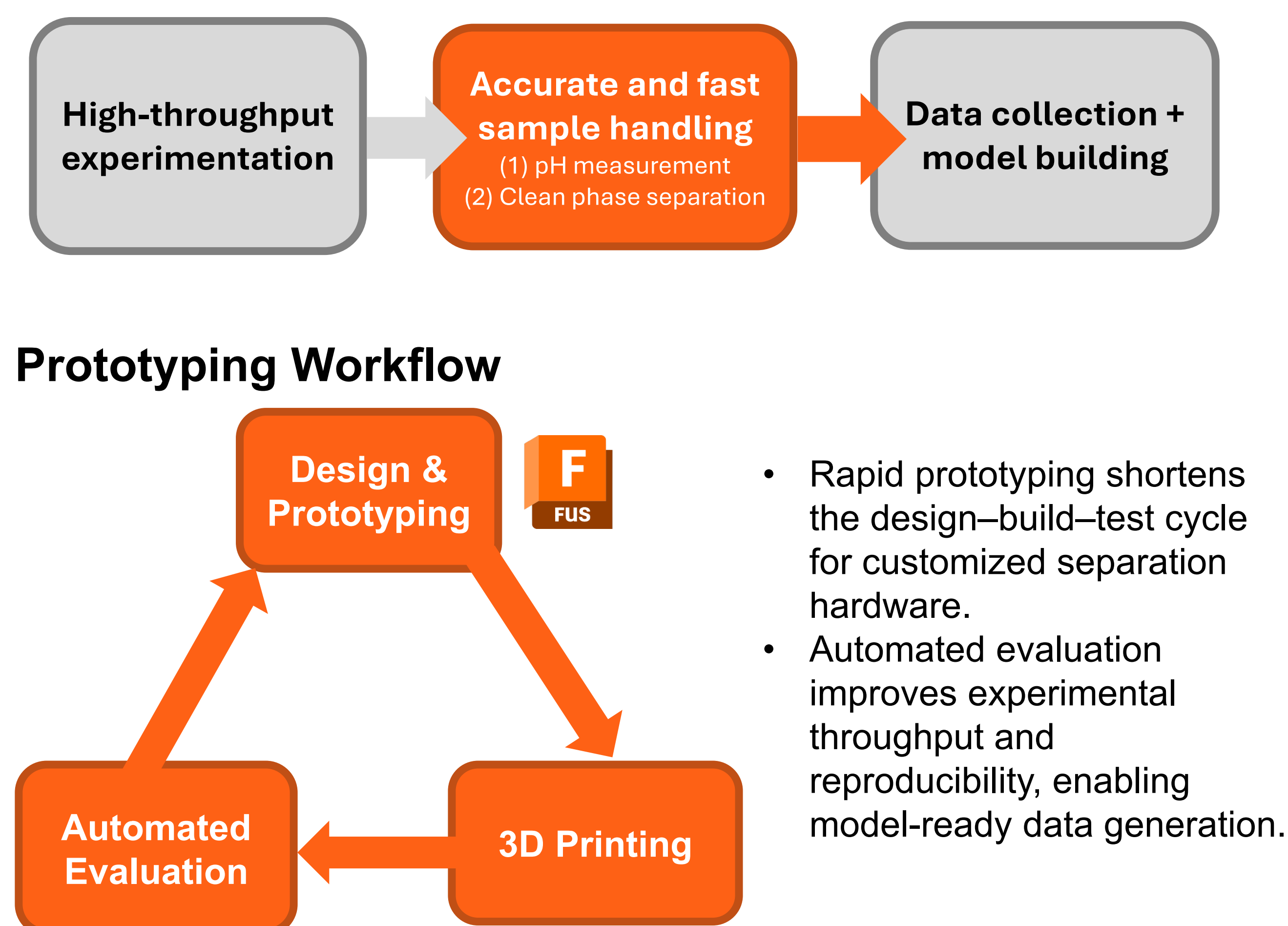
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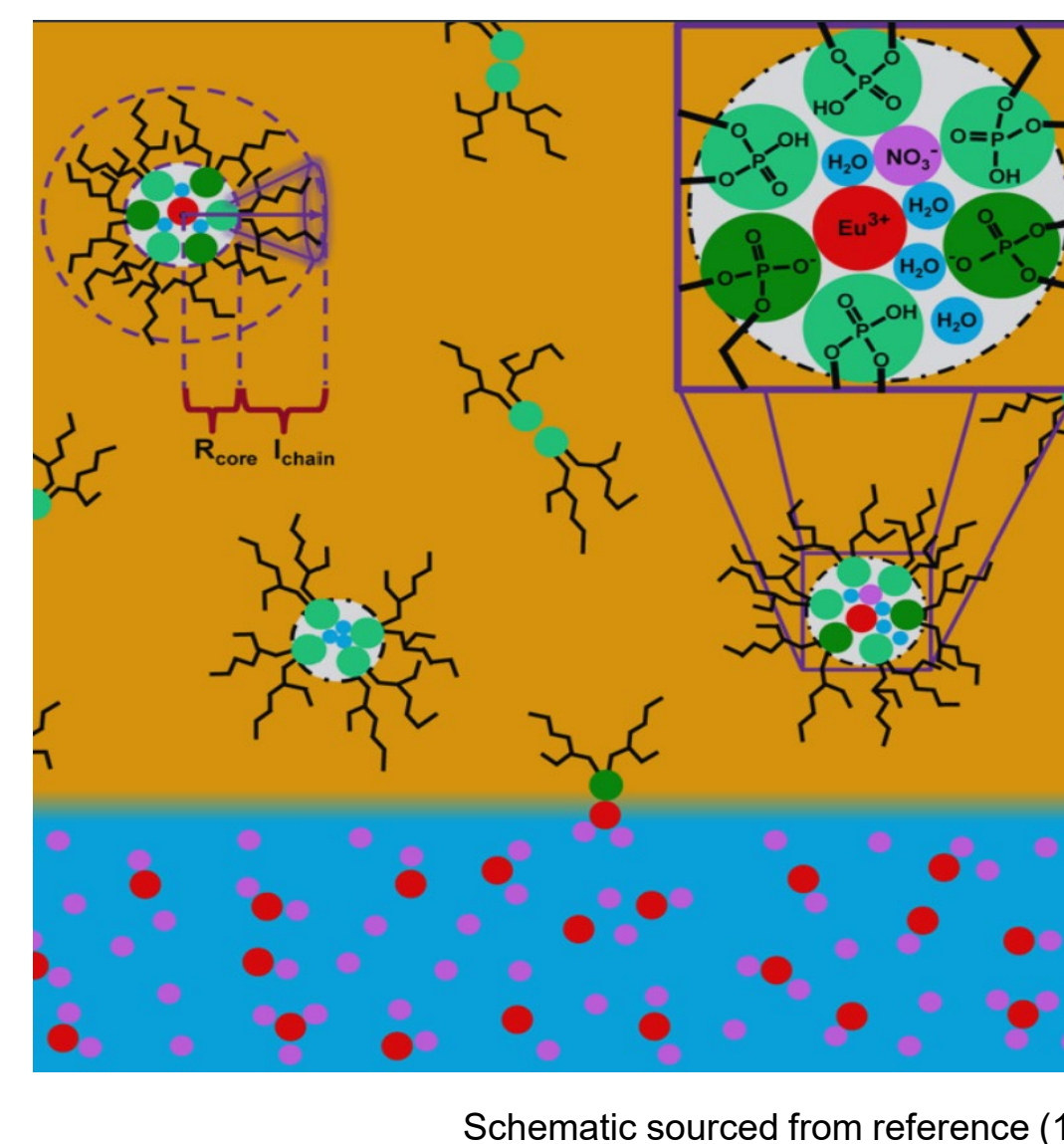
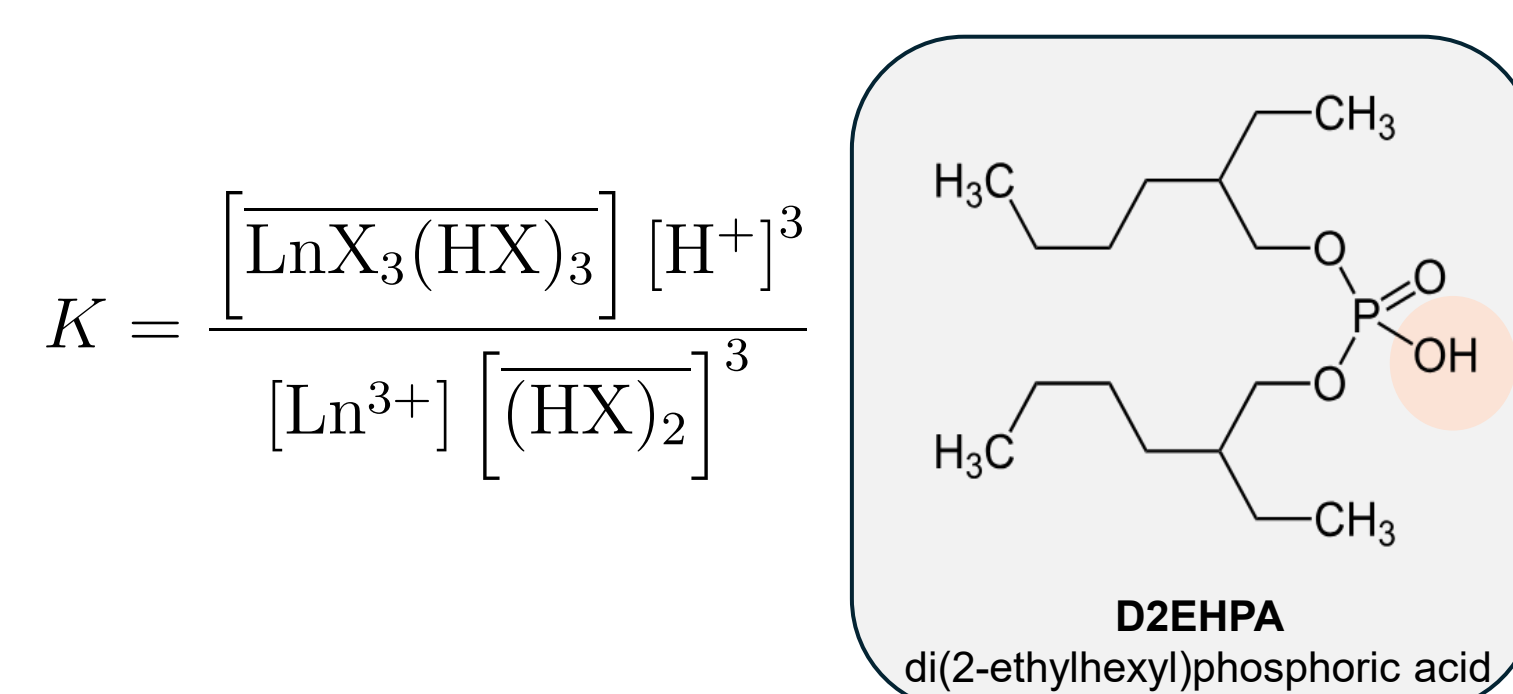
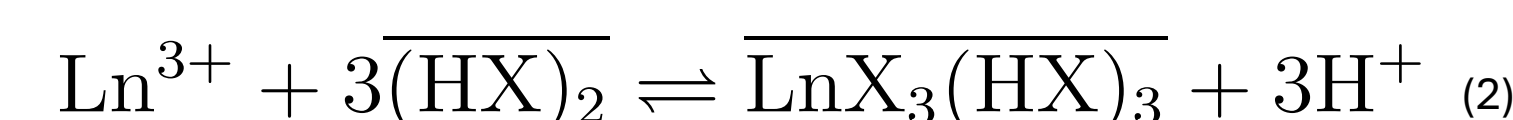
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Background and Motivation

Bottlenecks in the Critical Materials Separation Experimental Workflow



Importance of pH in Separations



- D2EHPA extracts Ln^{3+} through a pH-dependent cation-exchange reaction that releases H^+ into the aqueous phase.
- Many extractants exhibit pH-dependent selectivity toward different species
- Reproducible pH measurements can provide process information across many experimental conditions.

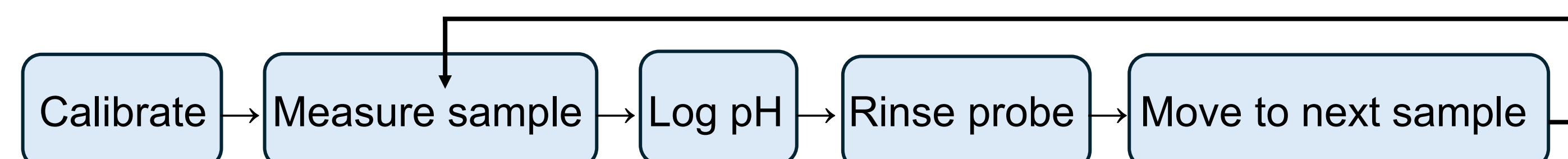
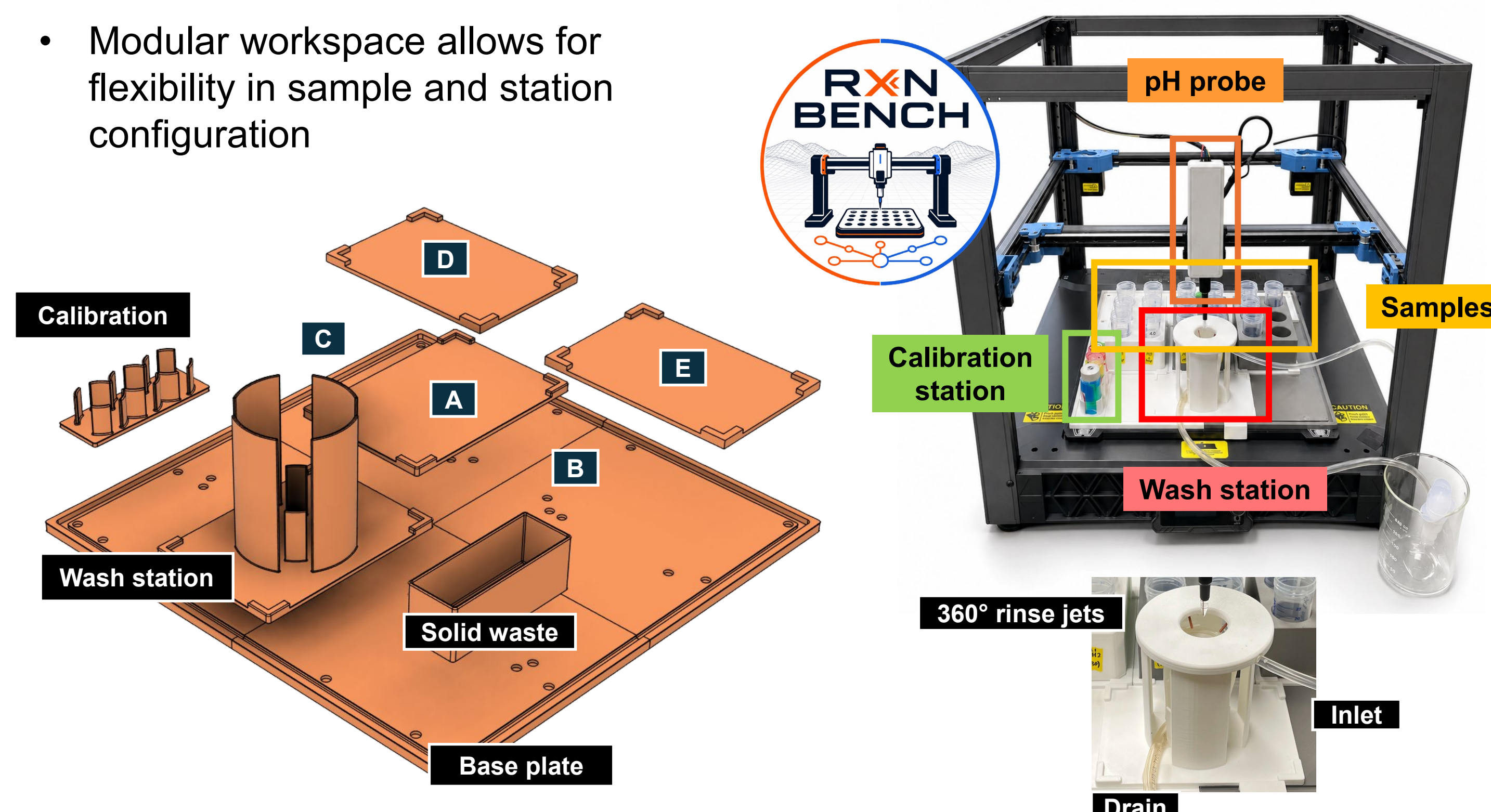
References

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- (4) Menzel, F., Klein, T., Ziegler, T. & Neumaier, J. M. 3D-printed PEEK reactors and development of a complete continuous flow system for chemical synthesis. *React. Chem. Eng.* **5**, 1300–1310 (2020).

Batch Separation

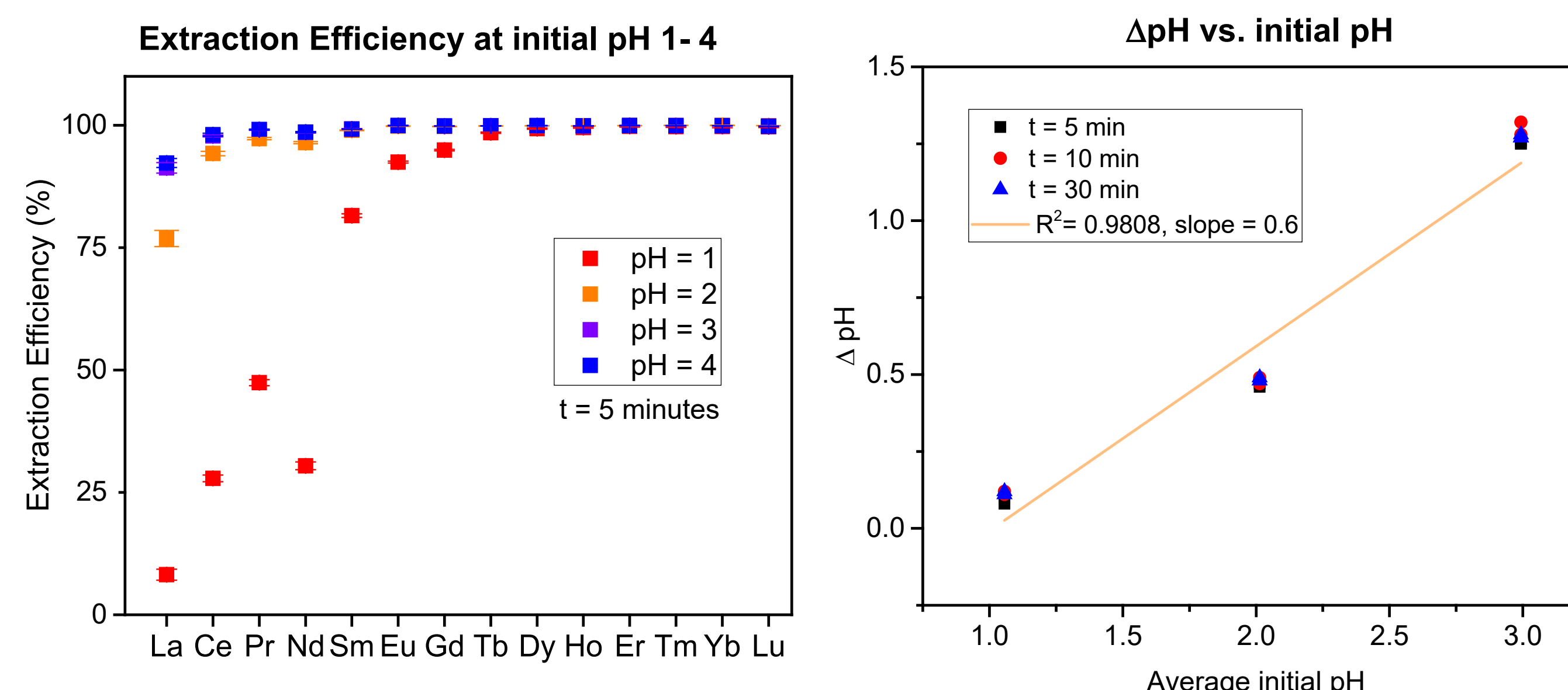
Rxn Bench: Automated pH Sampling Platform

- Modular workspace allows for flexibility in sample and station configuration



Rxn Bench assisted Extraction

- Batch extraction of a 7 mM total lanthanide mixture (0.5 mM each of 14 Ln^{3+} species) with 0.10 M D2EHPA in dodecane
- Rxn Bench measured and automatically logged pH for 40 samples spanning initial pH and extraction-time conditions.



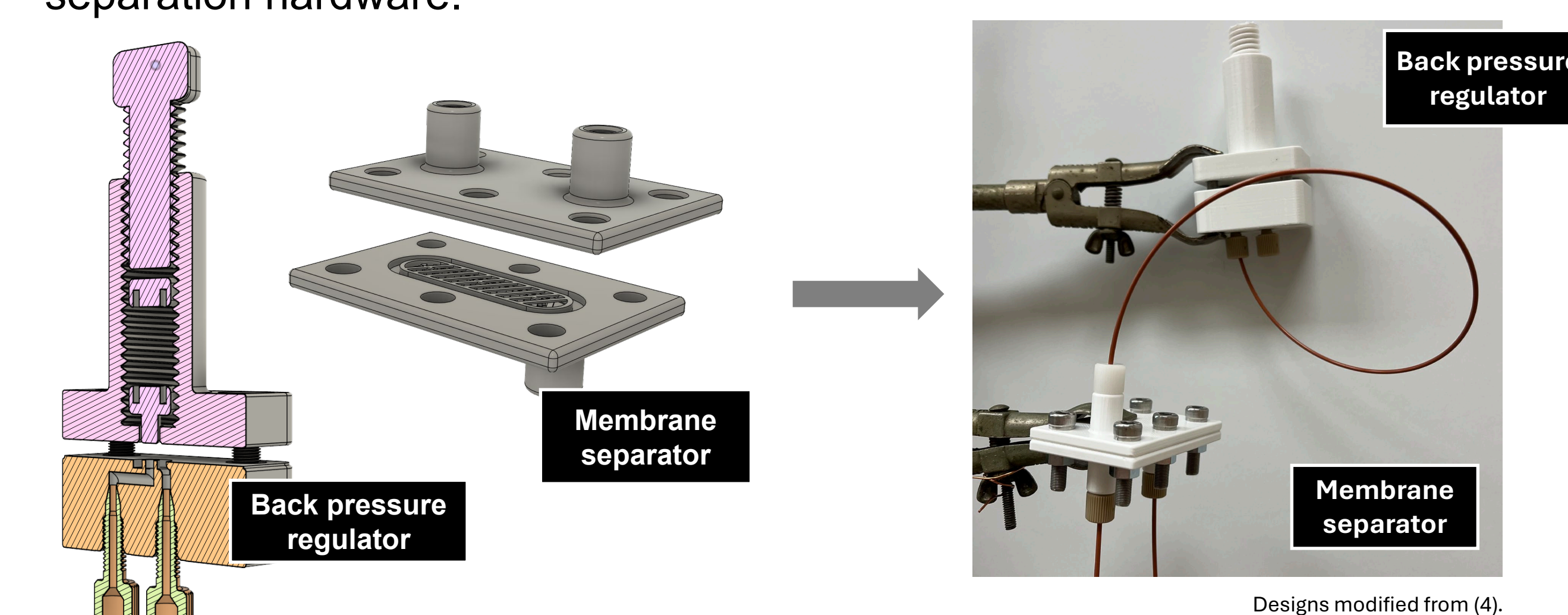
pH measurement approaches

Approach	Data quality	Throughput	Cost
Manual	Operator-dependent; manual logging	~ 24 samples/h	Labor intensive
Commercial automation	Reproducible	High	Typically >\$10,000 ⁽³⁾
Rxn Bench	Automated, reproducible logging	~48 samples/hour Scalable	<\$1000

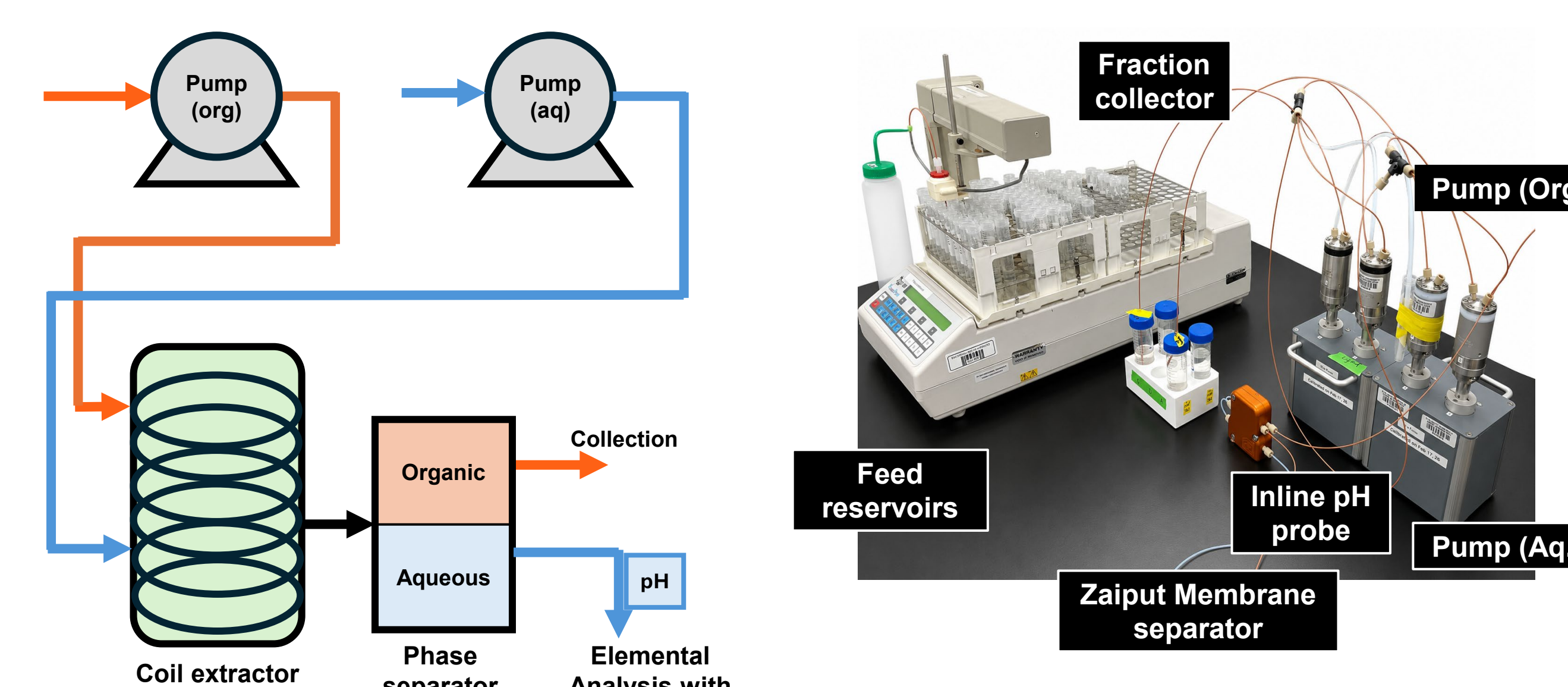
Flow Separation

Membrane Phase Separation

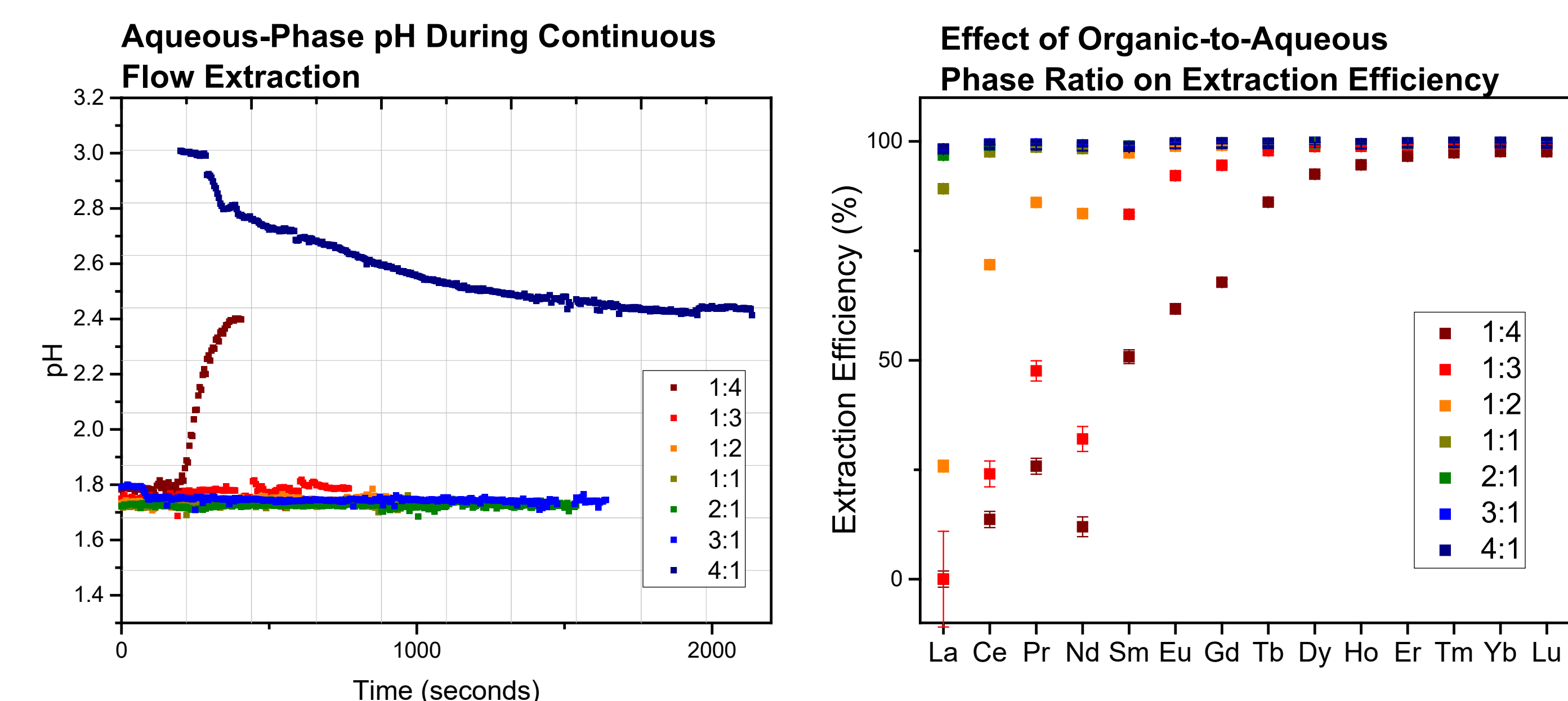
- A 3D-printed membrane separator and back-pressure regulator were adapted from an existing design for continuous organic-aqueous phase separation.
- Components can be fabricated and assembled in-house in ~5 hours. Rapid fabrication enables low-cost replacement and iterative modification of flow-separation hardware.



Continuous Extraction and Phase-Separation Workflow



- Extraction of a 7 mM total lanthanide mixture (0.5 mM each of 14 Ln^{3+} species) with 0.10 M D2EHPA in dodecane, initial aqueous pH = 3.
- Organic-to-aqueous phase volume ratio (O:A) was varied from 1:4 to 4:1



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